Plagiarism Scan Report

2% Plagiarism	
2% Exact Match	
0% Partial Match	
98% Unique	
Words	1000
Characters	7385
Sentences	335
Paragraphs	312
Read Time	23 minute(s)
Speak Time	31 minute(s)

Content Checked For Plagiarism

CSA0911 — Programming in Java
Smart Hospital Patient Appointment, Pharmacy Inventory and Alert Notification System

Submitted in the partial fulfilment for the Course of

CSA0911 PROGRAMMING IN JAVA

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by
AL NAJEEB 192424325
Kaviya Kumara 192425055
Sri vardhan 192325006
Venkata Anirudh 192572110

Under the Supervision of
Dr. R. Thalapathi Rajasekaran

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences Chennai-602105

AUGUST-2026

INTRODUCTION
The Smart Health Authority discussed in the problem statement requires the design and implementation of a menu-driven Java application for the automated management of a smart hospital. The application features patients scheduling appointments with doctors, dispensing of medicines from the inventory upon prescriptions, and sending out reminders for appointments as well as notifications when stocks of medicines

are low. The paper presents the team's design, implementation, testing, and analysis of the application which successfully demonstrates the use of object-oriented programming, Java Collection Framework, exception handling, and multithreading concepts.

The health management system designed and implemented by the team consists of six core entities – Patient, Doctor, Appointment, Medicine, Inventory, and Notification. Patient, Doctor, and Medicine classes encapsulate their data with appropriate access modifiers while inheritance is demonstrated through categories of patients and types of notifications. Instances of the Appointment, Inventory, and Notification classes are stored in ArrayList, HashSet, and HashMap objects using generic types and iterator objects. The application also demonstrates the use of three user-defined exceptions including InvalidIDException, DuplicateBookingException and OutOfStockException as well as the use of multithreading to demonstrate concurrent booking of appointments and sending of notifications. The two threads are synchronized using synchronized methods and wait/notify mechanism.

The Smart Health Authority discussed in the problem statement requires the design and implementation of a menu-driven Java application for the automated management of a smart hospital. The application features patients scheduling appointments with doctors, dispensing of medicines from the inventory upon prescriptions, and sending out reminders for appointments as well as notifications when stocks of medicines are low. The paper presents the team's design, implementation, testing, and analysis of the application which successfully demonstrates the use of object-oriented programming, Java Collection Framework, exception handling, and multithreading concepts.The health management system designed and implemented by the team consists of six core entities – Patient, Doctor, Appointment, Medicine, Inventory, and Notification. Patient.

1. Problem Statement and Problem Formulation

1.1 Problem Statement

Design, implement and evaluate a menu driven Java application for Smart Hospital Patient Appointment, Pharmacy Inventory and Alert Notification System. Modern hospitals are faced with numerous challenges as they strive to provide effective coordination among patient registration, doctor's availability, appointment scheduling, medicine inventory and timely patient notification. This can prove to be a challenge in cases where manual processes are involved, resulting in scheduling conflicts, errors in medicine stock records, delayed patient notification among other shortcomings.

The proposed Java application will be used to automate hospital operations including patient registration and appointment scheduling with doctor's availability. In addition, the pharmacy component of the system will be responsible for tracking medicine inventory and issuing alerts when medicine stocks are low. In cases where doctors have no further available appointment slots, patients will be placed in a waiting list and notified when an appropriate slot opens.

1.2 Problem Formulation

The problem has been divided into several sub-problems as described below:

(i) Patient and Appointment Management: This component deals with patient and doctor's registration, appointment booking, cancellation, search and record management.

The component will ensure doctor's schedules are managed effectively, for example, through automated waiting list in cases where patient appointments are declined due to lack of available slots.

(ii) Pharmacy Inventory Management: This component deals with available medicine stock management, recording used medicine, dosage management and patient alerting in cases where medicine stocks are low.

1.3 Objective and Expected Outcome

Some of the objectives include developing well-designed Java classes with encapsulation, inheritance and other object oriented programming concepts, implementing efficient data storage and retrieval mechanisms using Java Collection Framework including generics and iterators, and exception handling mechanisms including built in and user defined exceptions to prevent abnormal termination of the java application.

The expected outcome of the study is a fully working menu driven java application that is efficient in managing hospital appointments and pharmacy inventory. The application will be able to perform patient and doctor registration, appointment booking and cancellation, as well as waiting list management.

2. Requirements, Constraints and Assumptions

2.1 Functional Requirements

FR1 – Registration and management of patients (General, Senior, Emergency) and doctors with corresponding information

FR2 – Booking, cancellation, and listing of appointments; maintenance of a waitlist for fully booked doctors; automatic promotion of waitlisted patients upon cancellation

FR3 – Search and update of patient and medicine records by ID or name

FR4 – Tracking of pharmacy inventory; dispensing of medicines and decrementing of stock upon prescription; automatic raising of low-stock alerts

2.2 Technical Constraints

TC1 – Language: Java SE 17 or higher

TC2 – Collection Framework: ArrayList, HashSet, HashMap, Hashtable; generics and Iterator / ListIterator compulsory

TC3 – Concurrency: minimum two concurrent threads; synchronised methods and wait() / notify() inter-thread communication mandatory

TC4 – Exception handling: at least three user-defined exception classes; no abrupt program termination on invalid input

2.3 Assumptions

A1 – All data is held in memory; no need for file or database persistence

A2 – Each doctor has a fixed maximum appointment capacity per session

A3 – Thread.sleep() is used to simulate real-time notification dispatch latency

A4 – Patient IDs and medicine IDs follow the format P001–P999 and M001–M999 respectively

3. Application of Relevant Course Knowledge

3.1 Object Oriented Principles — CO1

Encapsulation: All fields of each entity are declared private and can only be accessed or modified via public getters, setters, and constructors. This prevents direct modification of the encapsulated data and enforces the principle of data hiding. Inheritance: Patient is an abstract superclass from which GeneralPatient, SeniorPatient, and EmergencyPatient are derived. Notification is the base class for AppointmentNotification and StockAlertNotification. Each subclass overrides type-specific

Matched Source

Similarity 2%

Title:Comment on "Dihydromyricetin improves DSS-induced ...

by RD Dhanushya — Saveetha College of Pharmacy, Saveetha Institute of Medical and Technical Sciences, Chennai 602105, India. Electronic address: ramaiyan ...Read more

https://google.com/goto?url=CAESYQHrOzAVNBvnaFnVpzq_yCAnPUyrsALcvIMx3VtI1wRVSIktamQ5s-s3dCRkj-v5XOHu2J1-JoHwCrZt2znl-xm8f6_33UDxs-IuRQzqMMYFzFqWgUiJPKo_Lsg7yvh7XiA